

Solution

(a) Proof that the mean of distribution $\hat{p}(x)$ equals the sample mean

The mean of distribution $\hat{p}(x)$ is defined as:

$$\hat{\mu} = \mathbb{E}_{\hat{p}}[x] = \int x \hat{p}(x) dx$$

Substituting $\hat{p}(x) = \frac{1}{n} \sum_{i=1}^n \tilde{k}(x - x_i)$:

$$\hat{\mu} = \int x \left(\frac{1}{n} \sum_{i=1}^n \tilde{k}(x - x_i) \right) dx$$

Interchanging the order of integration and summation:

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n \int x \tilde{k}(x - x_i) dx$$

For each integral, let $y = x - x_i$, then $x = y + x_i$, $dx = dy$:

$$\int x \tilde{k}(x - x_i) dx = \int (y + x_i) \tilde{k}(y) dy = \int y \tilde{k}(y) dy + x_i \int \tilde{k}(y) dy$$

By the properties of the kernel function: $\int \tilde{k}(y) dy = 1$ and $\mathbb{E}_{\tilde{k}}[y] = \int y \tilde{k}(y) dy = 0$, therefore:

$$\int x \tilde{k}(x - x_i) dx = 0 + x_i \cdot 1 = x_i$$

Substituting back:

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i$$

(b) Proof that the covariance of distribution $\hat{p}(x)$ has the given form

The covariance of distribution $\hat{p}(x)$ is defined as:

$$\hat{\Sigma} = \text{cov}_{\hat{p}}(x) = \int \hat{p}(x) (x - \hat{\mu})(x - \hat{\mu})^T dx$$

Substituting $\hat{p}(x)$:

$$\hat{\Sigma} = \int (x - \hat{\mu})(x - \hat{\mu})^T \left(\frac{1}{n} \sum_{i=1}^n \tilde{k}(x - x_i) \right) dx = \frac{1}{n} \sum_{i=1}^n \int (x - \hat{\mu})(x - \hat{\mu})^T \tilde{k}(x - x_i) dx$$

For each integral, let $y = x - x_i$, then $x = y + x_i$, $dx = dy$:

$$\int (x - \hat{\mu})(x - \hat{\mu})^T \tilde{k}(x - x_i) dx = \int (y + x_i - \hat{\mu})(y + x_i - \hat{\mu})^T \tilde{k}(y) dy$$

Expanding the integrand:

$$(y + x_i - \hat{\mu})(y + x_i - \hat{\mu})^T = yy^T + y(x_i - \hat{\mu})^T + (x_i - \hat{\mu})y^T + (x_i - \hat{\mu})(x_i - \hat{\mu})^T$$

Integrating term by term:

- $\int yy^T \tilde{k}(y) dy = H$ (by the definition of kernel covariance)
- $\int y(x_i - \hat{\mu})^T \tilde{k}(y) dy = \left(\int y \tilde{k}(y) dy \right) (x_i - \hat{\mu})^T = 0 \cdot (x_i - \hat{\mu})^T = 0$
- $\int (x_i - \hat{\mu})y^T \tilde{k}(y) dy = (x_i - \hat{\mu}) \left(\int y^T \tilde{k}(y) dy \right) = (x_i - \hat{\mu}) \cdot 0 = 0$
- $\int (x_i - \hat{\mu})(x_i - \hat{\mu})^T \tilde{k}(y) dy = (x_i - \hat{\mu})(x_i - \hat{\mu})^T \int \tilde{k}(y) dy = (x_i - \hat{\mu})(x_i - \hat{\mu})^T$

Therefore, the integral result is:

$$\int (x - \hat{\mu})(x - \hat{\mu})^T \tilde{k}(x - x_i) dx = H + (x_i - \hat{\mu})(x_i - \hat{\mu})^T$$

Substituting into $\hat{\Sigma}$:

$$\hat{\Sigma} = \frac{1}{n} \sum_{i=1}^n [H + (x_i - \hat{\mu})(x_i - \hat{\mu})^T] = H + \frac{1}{n} \sum_{i=1}^n (x_i - \hat{\mu})(x_i - \hat{\mu})^T$$

(c) Analysis of properties and their relationship with bias

The above results indicate that the kernel density estimate $\hat{p}(x)$ has the following properties:

1. **Mean property:** From (a), the mean $\hat{\mu}$ of $\hat{p}(x)$ equals the sample mean, which means that when estimating the population mean, the kernel density estimate is first-order unbiased. That is, as an estimator of the mean, $\hat{\mu}$ is an unbiased estimator of the true mean.
2. **Covariance property:** From (b), the covariance $\hat{\Sigma}$ of $\hat{p}(x)$ consists of two terms: the covariance H of the kernel function and the sample covariance. This indicates that when estimating the population covariance, the kernel density estimate is biased, with bias equal to H . Specifically, $\mathbb{E}[\hat{\Sigma}] = H + \Sigma$ (where Σ is the true covariance), so the bias is H .
3. **Relationship with the bias of the kernel density estimator:** These properties reflect the smoothing effect of kernel density estimation. The use of a kernel function introduces smoothing,

causing the estimated distribution $\hat{p}(x)$ to be more dispersed than the empirical distribution. In density estimation itself, the bias typically depends on the bandwidth (implicit in H): when the bandwidth is large, the degree of smoothing is high, the variance decreases but the bias increases (especially in point estimation of the density function); when the bandwidth is small, the degree of smoothing is low, the bias decreases but the variance increases. Here, from the perspective of moments, it is shown that the kernel density estimate is unbiased for the first moment but biased for the second moment, and the bias is directly determined by the kernel covariance H . Therefore, in practical applications, it is necessary to balance the bandwidth selection to control the overall bias and variance.

In summary, the kernel density estimate is unbiased for mean estimation but has positive bias for variance estimation, which stems from the smoothing characteristics of the kernel function.